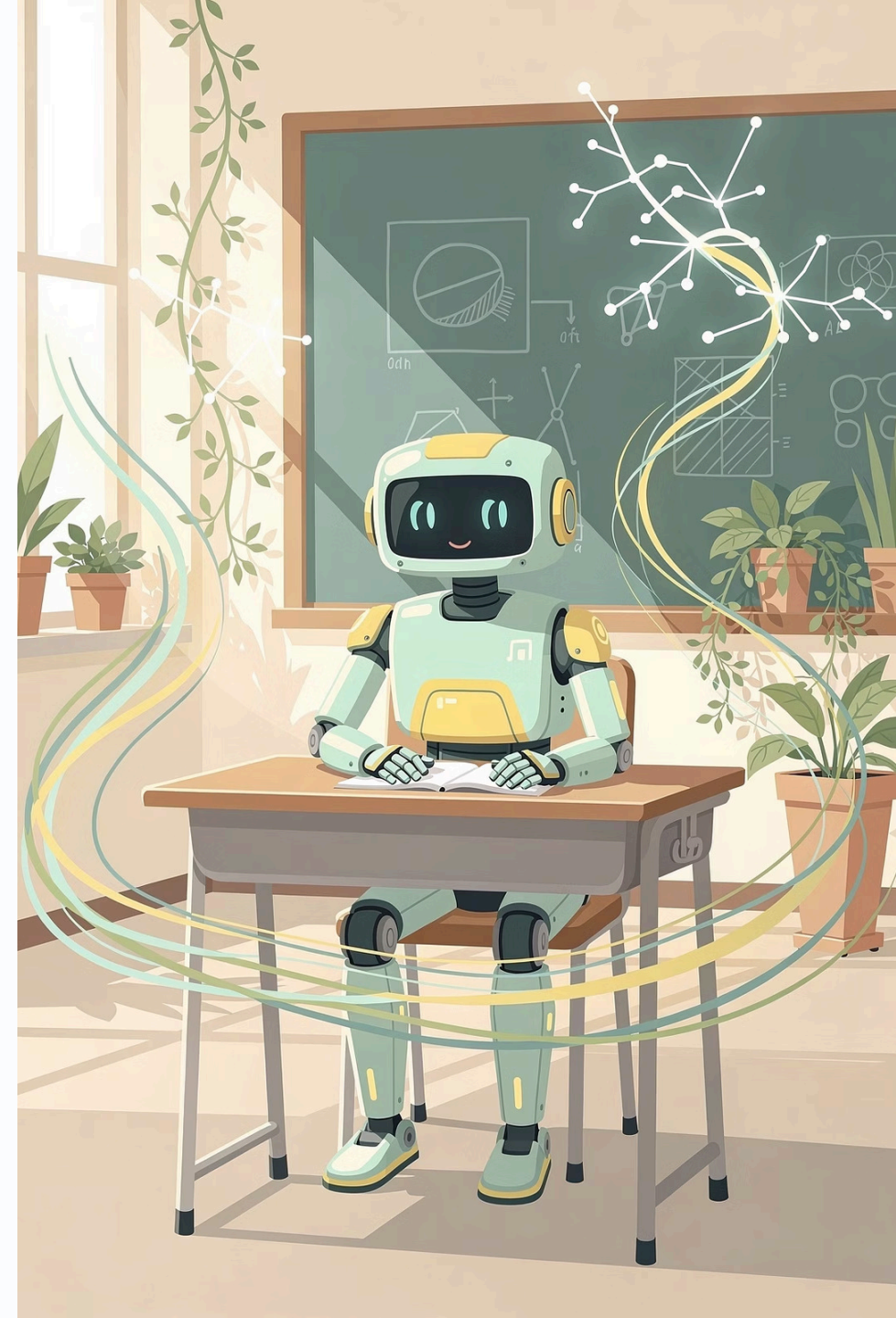


Session 3: What is Machine Learning?

You've been hearing "Machine Learning" everywhere. Today, we finally find out what it really means - and how computers actually learn.

Spoiler: it's not magic. It's patterns.



Objectives

By the end of this session, you will be able to:

01

Define Machine Learning

Say what Machine Learning is in your own words

02

Compare Systems

Tell the difference between rules-based and learning-based systems

03

Spot the Limits

Explain why rules can't solve every problem

04

Think Like AI

Spot patterns the way AI does

05

Train a Real AI

See a real AI model learn live with Google Teachable Machine!



Recap & Homework Share

Welcome back! Let's start with your **AI Detective** work.

Your Investigation

- Which AI did you investigate?
- What data do you think it was trained on?
- What does it predict?

Quick check from last session - can you say the AI Recipe?



Data

The examples AI
learns from



Algorithm

The method for
learning



Training

The learning process



Model

The finished AI



Prediction

The AI's answer

Warm-Up - How Did You Learn?

Let's think about how **YOU** learn. How did you learn what a dog looks like?

The Rulebook Way 📋

Did someone hand you a rulebook? *"A dog has 4 legs, fur, and barks..."*

The Experience Way 👁️

Or did you just **SEE** a lot of dogs and figure it out over time?

❏ **Big Question:** Could computers learn the same way you did?



What is Machine Learning?

Machine Learning (ML) = when computers learn patterns from examples instead of following rules a person wrote.



Old Way (Rules)

Programmer figures out the rules

Computer follows the rules exactly

Same answer every time



New Way (Learning)

Computer figures out the patterns from examples

Computer finds patterns on its own

Gets better the more it sees



ML is the second way. And it's what powers almost every "smart" app you use today.

learning from examples



$$sx = \frac{2}{2} 8 2 \cdot 2 ()$$

$$1 \cdot = \cdot p^1 x + 6$$

$$4 + = \cdot + x \frac{1}{2} + ^2$$

$$5x = \cdot x x + ^3$$

$$2x = x^2 x + ^2$$

$$\frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

$$y = \frac{2x = 13}{2^{10}} \frac{3}{2}$$

Rules - The Old Way

Most of the computer stuff you use follows rules a person wrote.

These are **reliable** - they never surprise you. But they can't learn.



Calculator

"If you press $2 + 2$, show 4." Always.



Traffic Light

"Green for 30 seconds, then yellow for 3, then red for 30." Always.



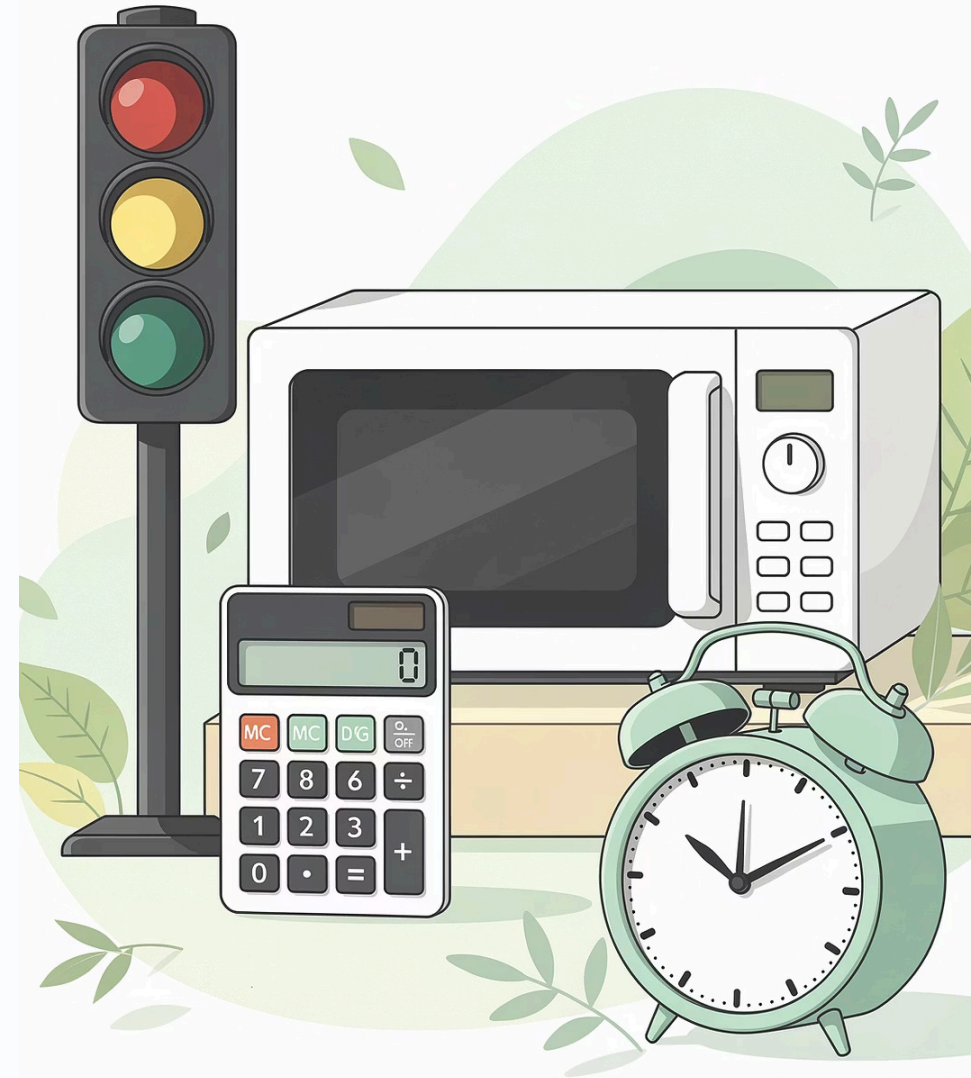
Microwave

"If popcorn button is pressed, run for 2 minutes." Always.



Alarm Clock

"If time = 7:00 AM, ring." Always.





Why Rules Fail

The Spam Filter Problem

Imagine writing rules for a spam filter:

- Rule 1: If email says "FREE MONEY!!!" → spam
- Rule 2: If email says "You won a prize" → spam

But then bad guys get sneaky:

- They write "F R E E M O N E Y" with spaces
- They write "Congratulations winner!"
- They pretend to be your friend

☐ This is why we need Machine Learning.

Mini Activity 1 - Rules or Learning?

For each system below, decide: is it **RULES** (old way) or **LEARNING** (new way)?

System	Your Guess
 Calculator	?
 YouTube recommendations	?
 Traffic light	?
 Gmail spam filter	?
 Microwave preset buttons	?
 Face unlock on a phone	?
 Netflix suggestions	?
 Snapchat filters	

Mini Activity 1 - Answers

System	Answer
 Calculator	Rules
 YouTube recommendations	Learning
 Traffic light	Rules
 Gmail spam filter	Learning
 Microwave preset buttons	Rules
 Face unlock on a phone	Learning
 Netflix suggestions	Learning
 Snapchat filters	Learning

 **Easy rule:** If it gets better over time → Learning. If it does the exact same thing every time → Rules.

What Are Patterns?

Patterns = things that repeat. Your brain finds them all the time - and so does AI.

Dark clouds 🌧️

"It's going to rain" - you read the sky without thinking about it.

Teacher's hint 📝

"This might be on the test" - you know to pay attention.

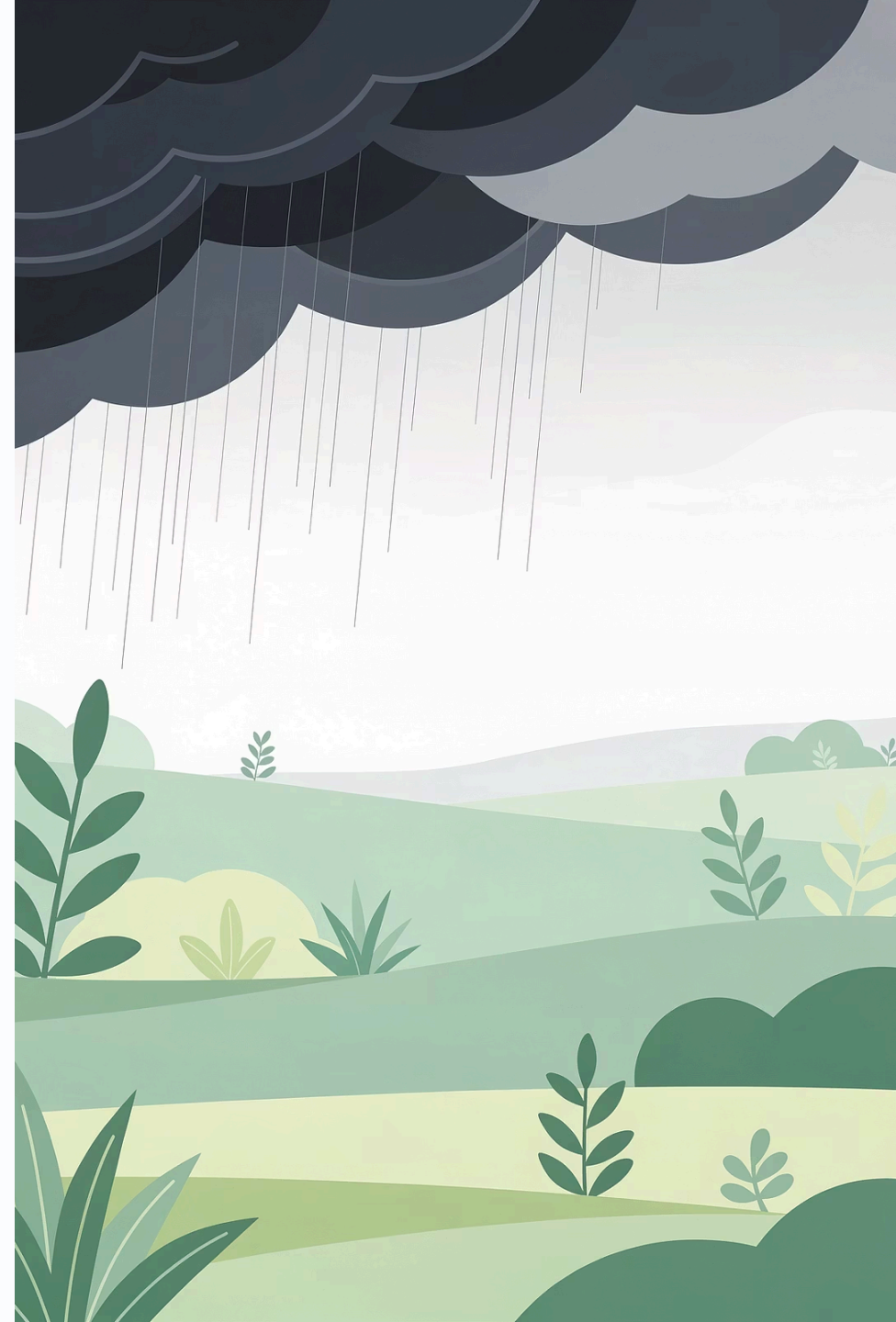
Mom is humming 😊

"She's in a good mood" - you read the signal instantly.

Dog runs to door 🐕

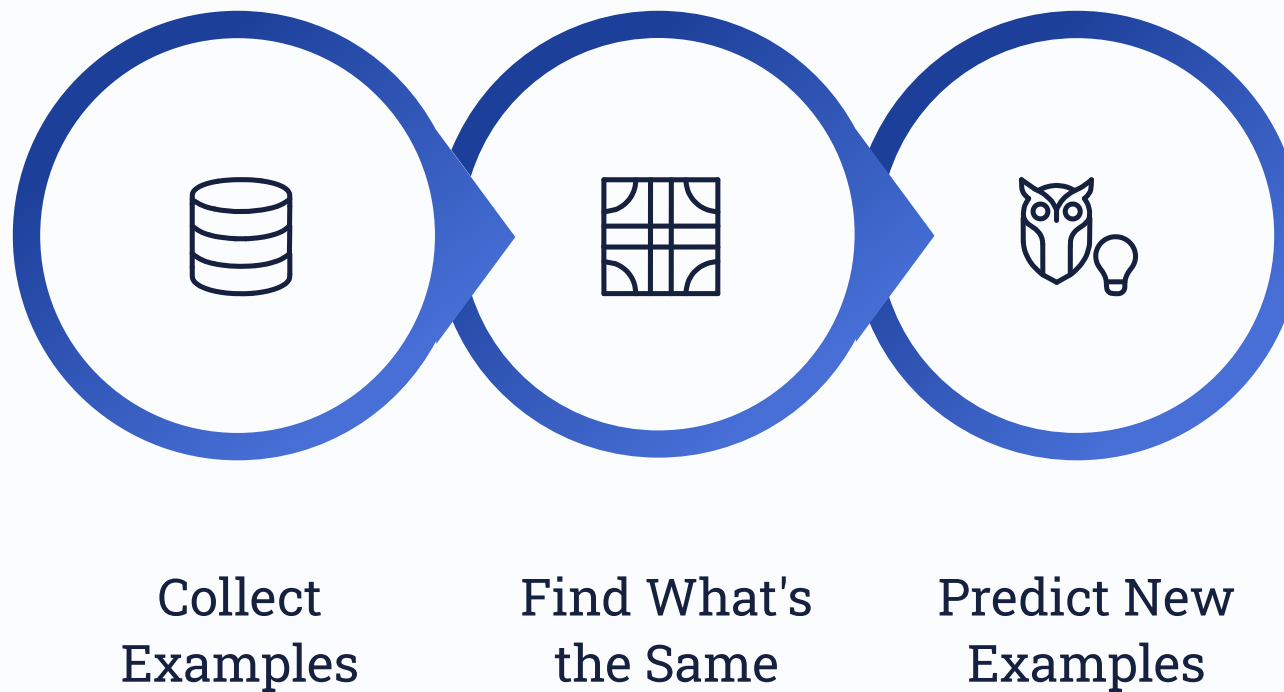
"Someone's home!" - your dog spotted the pattern before you did.

✅ **Spotting patterns IS learning.** That's exactly what Machine Learning does - but with WAY more examples than your brain could ever hold.



How Does AI Find Patterns?

Three simple steps - that's all it takes:



That's it. That's how ML works - collect, find patterns, predict. Simple in concept, powerful in practice.

Mini Activity 2 - Spot the Pattern

You're going to do what AI does. Find the pattern and predict what comes next!

Number Patterns

2, 4, 6, 8, ____

What comes next?

5, 10, 15, 20, ____

What comes next?

Shape Patterns

○, △, ○, △, ____

What comes next?

●, ●, ●, ●, ●, ____

What comes next?

Trickier Patterns

Dog, Cat, Dog, Cat, Dog, ____

What comes next?

Sunny, Sunny, Rainy, Sunny, Sunny,
Rainy, ____

What comes next?

📋 You just did Machine Learning! You saw examples, found what was the same, and predicted what came next. That's **exactly** what ML does - just with millions of examples and way faster.

Mini Activity 2 - Answers

Number Patterns

2, 4, 6, 8, ----

10

5, 10, 15, 20, ----

25

Shape Patterns

○, △, ○, △, ----

○

●, ●, ●, ●, ●, ----

●

Trickier Patterns

Dog, Cat, Dog, Cat, Dog, ----

Cat

Sunny, Sunny, Rainy, Sunny,
Sunny, Rainy, ----

Sunny



ML in Your Life Right Now

Remember the list of AI apps you made in Session 1? Surprise - **all of them are using Machine Learning**. That's what's behind the scenes:

TikTok / Instagram

ML learns what videos you watch fully → shows you more

Spotify

ML learns what songs you replay → makes your Discover Weekly

Google Maps

ML learns from millions of phones → predicts traffic

Snapchat Filters

ML learns where your eyes, nose, mouth are → puts filters on

Gmail Spam Filter

ML learns what spam looks like → blocks it

Hands-On Time

Step 1: Opening Google Teachable Machine

Time to see ML happen **LIVE**. Today's tool: **Google Teachable Machine** - a free tool that lets us train our OWN AI in just a few minutes.

We'll teach the AI to recognize 3 hand gestures:

✌️ Peace Sign

👋 High Five

✊️ Fist

- 📄 Today we'll watch and see how it works. In a future session, you'll build your own!

PEACE



HIGH FIVE

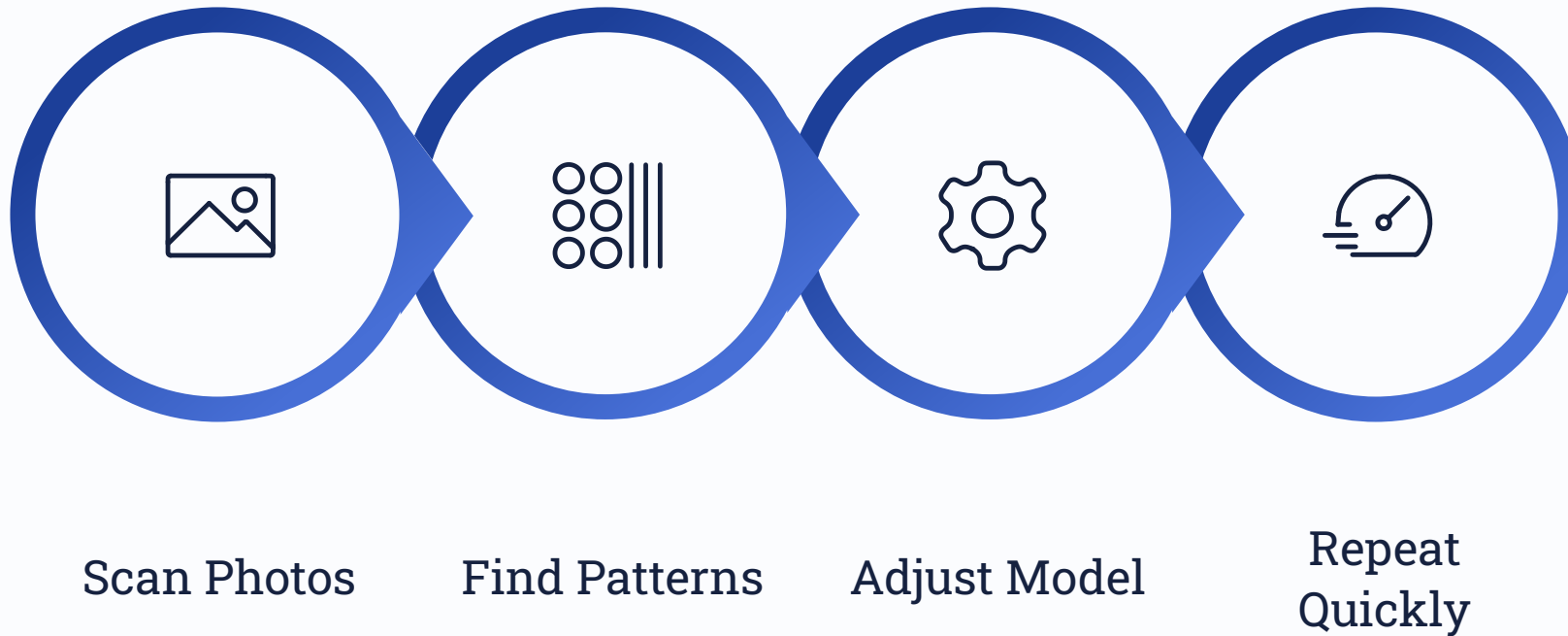


FIST

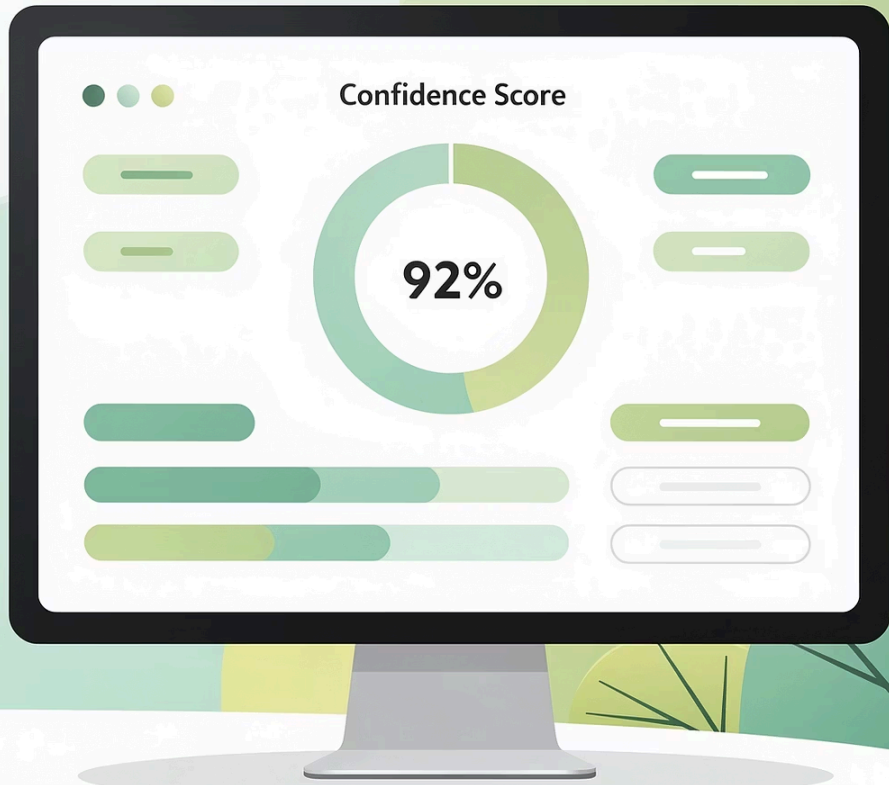


Step 2 - TRAINING Happens!

Click "**Train Model.**" Watch what happens:



✓ You just watched **TRAINING** happen live!



Step 3 - Test It! (PREDICTIONS!)

Now show the AI a **NEW gesture** - one it hasn't seen yet.

What You'll See

It shows a percentage - e.g., "**92% Peace Sign**"

That's its **confidence score**. That's a **PREDICTION**.

Try to Trick It!

- Show your hand from a totally new angle
- Try a gesture it DIDN'T learn (like "rock on" 🤘)
- What happens? Does it get confused?

⚠️ What does it mean if it guesses wrong? It means ML can only recognize patterns it was **TRAINED on**. Show it something new = it doesn't know what to do.

What Just Happened? The Full AI Recipe!

We just did the whole AI Recipe in 10 minutes!



DATA

The hand gesture photos we took



ALGORITHM

Teachable Machine's learning method



TRAINING

When we clicked "Train Model"



MODEL

The finished AI that can now recognize gestures



PREDICTION

Every guess it made with a confidence %

You just trained an AI. From scratch. In 10 minutes. 🎉

What We Learned Today

Machine Learning

Traditional Programming

Why Rules Fail

Pattern Recognition

Learning from Patterns

ML in Everyday Life

Your Homework - ML Detective

Find **3 apps or systems** you use that use Machine Learning. For each one, write down:

What is it?

YouTube, Spotify, Snapchat, etc.

What does it do?

Describe its main function

What patterns did it learn?

What data do you think it was trained on?

Examples to choose from:

- Spotify, Apple Music
- YouTube, Netflix, TikTok
- Snapchat, Instagram filters
- ChatGPT, Gemini
- Face Unlock on a phone
- Google Maps

✓ **Bonus:** Try to explain Machine Learning to someone at home - a parent, sibling, or friend.



Next Time - Session 4: Types of Machine Learning

Now that you know **WHAT** Machine Learning is, next time we'll find out the different ways machines can learn!



Supervised Learning

Learning with answer keys



Unsupervised Learning

Finding patterns alone



Reinforcement Learning

Learning by trial and error

 **Don't forget:** Bring your ML Hunt! 🕵️ We'll share at the start of Session 4.